DSA0602 DATA HANDLING AND VISUALIZATION | Class Activity 2: Relationship Workshop in R (35 Marks) | Slot D

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Scatterplots | Correlograms | PCA | Networks | Paired data | Bubble charts | Hexbin plots | 7 tasks x 5 marks | Tool: R (ggplot2 and the packages named in each question)

Instructions: Tool: R (ggplot2 and the packages named). Time: 3 hours. Total: 35 marks (7 tasks x 5). Every student does the same seven tasks; the data in the R code printed with each task are your own, so your numbers will differ from your neighbour's. Run each code block unchanged (do not edit the seed). Draw the chart asked for, then answer with specific values from YOUR output; generic statements earn no credit. Each chart needs a title, labelled axes with units and colour-blind-safe colours. Submit one R script, your charts and short answers as one file named RollNo_Name.

Relationship Workshop

Q1 [Scatterplots] (5 Marks) Scatterplot

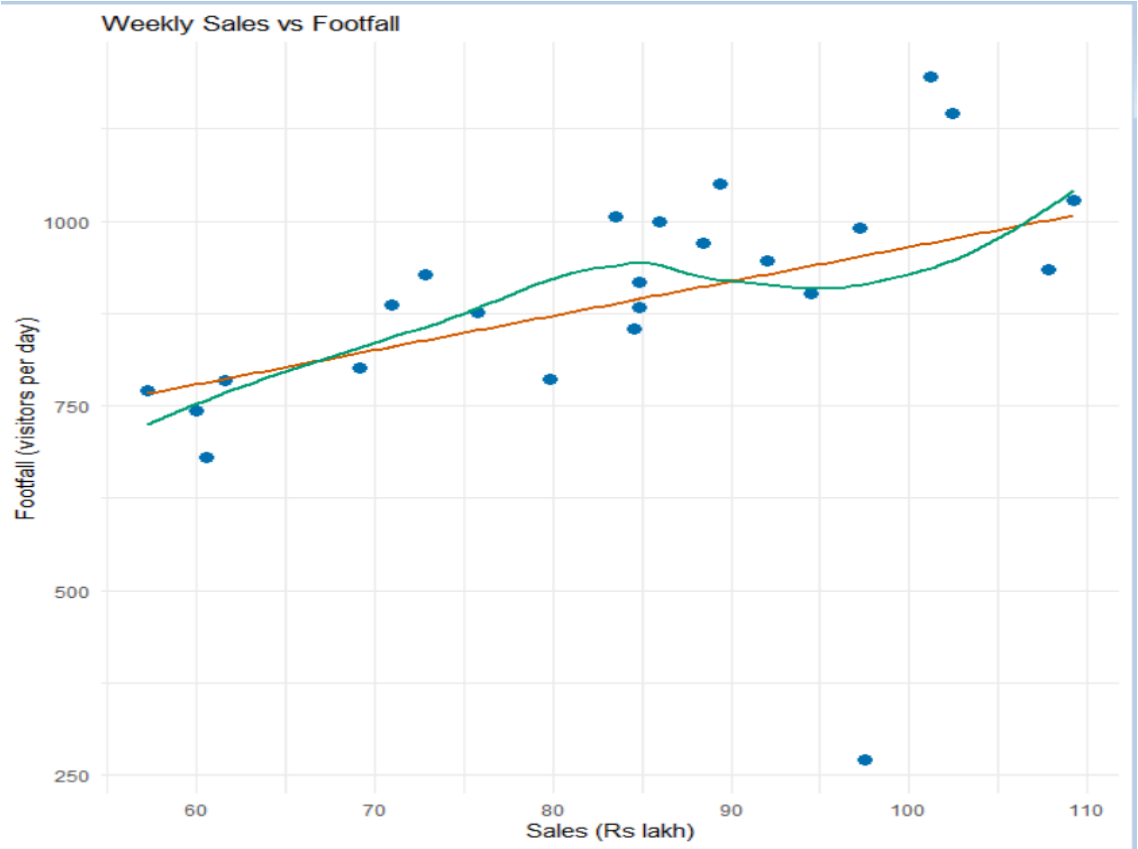
Study weekly sales (Rs lakh) against footfall (visitors per day) across 24 weekly records.

```
set.seed(1282053)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200.0 * (0.75 * z + 0.00 * z^2 + 0.41 * rnorm(n)), 0)
footfall[18] <- round(footfall[18] - 3.8 * 200.0, 0) # one record was keyed in wrongly
df <- data.frame(sales, footfall)
```

Tasks:

1. Draw a scatterplot of sales (x) against footfall (y) with a straight-line fit (geom_smooth(method = "lm")) and a loess curve; label axes with units. [1.5]

2. Report the correlation r and the slope of the line, and describe the strength and direction in one sentence. [1] 3. Flag the outlier: give its row number and values, using the largest standardised residual (rstandard) from the straight-line fit. [1] 4. Fit a quadratic model (y ~ x + I(x^2)) and compare R-squared with the straight line; state whether a linear or non-linear trend fits better. [1] 5. Give one recommendation for the Sales Director that follows from the chart. [0.5]



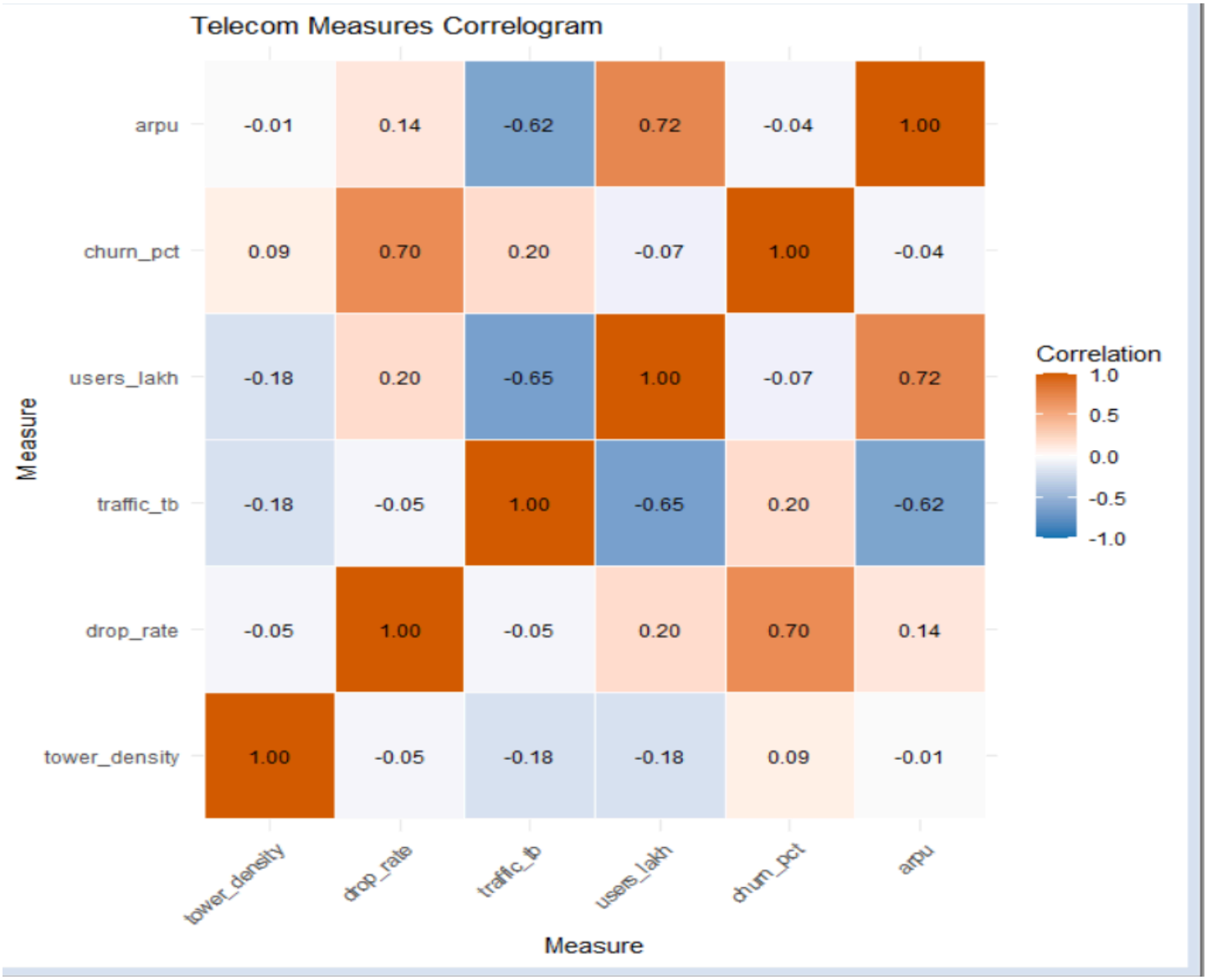
Q2 [Correlograms] (5 Marks) Correlogram (correlation matrix heatmap)

Study the six measures recorded for 30 circles-periods at a telecom firm. The data frame df holds all six.

```
set.seed(1282125)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(rnorm(n, 14.0, 4.0), 1)
drop_rate <- round(2.6 + 0.7 * (0.86 * f2 + 0.52 * rnorm(n)), 1)
traffic_tb <- round(320 + 70.0 * (-0.76 * f1 + 0.65 * rnorm(n)), 0)
users_lakh <- round(24.0 + 6.0 * (0.85 * f1 + 0.52 * rnorm(n)), 1)
churn_pct <- round(1.8 + 0.5 * (0.82 * f2 + 0.57 * rnorm(n)), 1)
arpu <- round(185 + 25.0 * (0.84 * f1 + 0.54 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb, users_lakh, churn_pct, arpu)
```

Tasks:

1. Compute cor(df), round to 2 decimals, and draw a correlogram (ggplot2 geom_tile with geom_text, or corrplot) with a diverging colour scale centred on 0. [1.5]
2. Identify the two most strongly correlated variable pairs (by absolute r) and report their r. [1.5]
3. Identify one surprising weak or negative correlation, report its r and explain in one sentence what it implies. [1] 4. Give one recommendation for the Network Head based on the matrix. [1]



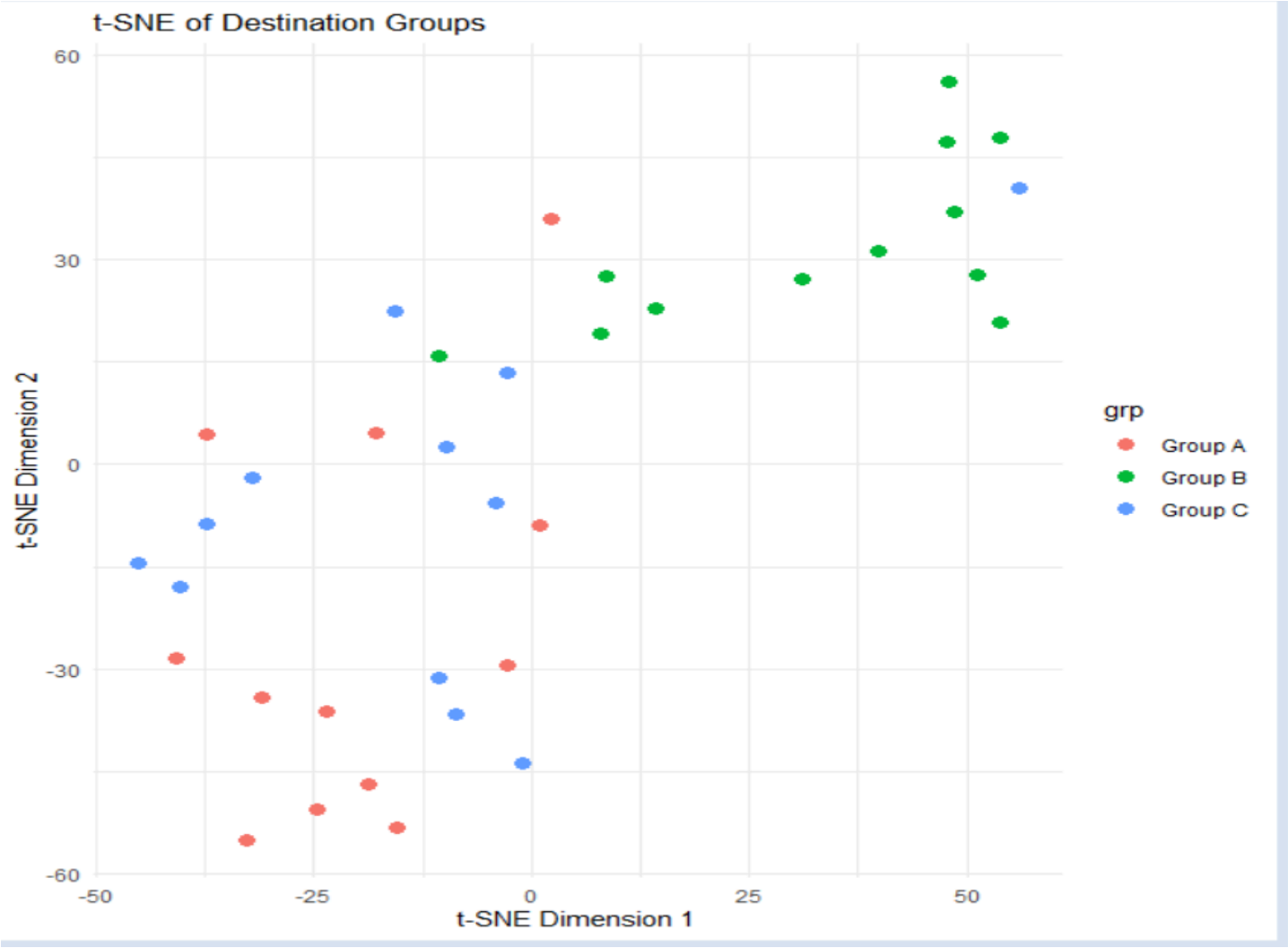
Q3 [Dimension Reduction (PCA, t-SNE, UMAP)] (5 Marks) Dimension Reduction (PCA and t-SNE/UMAP) Six measures were recorded for 36 destinations from three known groups (column grp).

```
set.seed(1282182)
n <- 36
```

```
grp <- rep(c("Group A", "Group C", "Group B"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120.0 + 30.0 * (rnorm(n) + 0.70 * g), 1)
spend_rs <- round(3200 + 600.0 * (rnorm(n) + 0.80 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.80 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.06 * g), 1)
occupancy_pct <- round(68.0 + 10.0 * (rnorm(n) + 0.08 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 1.00 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights, rating10, occupancy_pct, complaints)
```

Tasks:

1. Run PCA on the scaled data (prcomp(df, scale. = TRUE)). Report the percentage of variance for PC1 and PC2 and the number of components needed for at least 80%. [1.5]
2. Plot PC1 against PC2 coloured by grp, then report the variable with the largest absolute loading on PC1. [1]
3. Judge how well PCA separates the groups using the share of variance in each of PC1 and PC2 explained by grp (R-squared of lm(score ~ grp)). [1] 4. Make a t-SNE (Rtsne) or UMAP plot of the same data (install the package if needed) and explain the variance-versus-neighbourhood trade-off between it and PCA; give one recommendation for the analyst. [1.5]



Q4 [Network Graphs] (5 Marks) Network Graph

Study the connections (flows of parcels delivered) between 8 hubs. Each row of the data is a link with a weight. set.seed(1282246)

```
nodes <- c("Ahmedabad", "Pune", "Delhi", "Chennai", "Bengaluru", "Kolkata", "Hyderabad", "Mumbai")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
keep <- u < 0.25 | ((pairs[, 1] == "Ahmedabad" | pairs[, 2] == "Ahmedabad") & u < 0.91)
edges <- data.frame(from = pairs[keep, 1], to = pairs[keep, 2], weight = round(runif(sum(keep), 1, 20)))
```

Tasks:

1. Build the network with igraph (graph_from_data_frame). Draw it with node size proportional to degree and edge width proportional to weight, with node labels. [1.5]

2. Compute degree and betweenness for every node and name the most central node, justifying it with both numbers. [1.5] 3. Report the number of edges, the density (edge_density) and the number of connected components. [1]

4. Delete the most central node (delete_vertices) and report how many components remain; give one recommendation for the Operations Head. [1]

Q5 [Paired Data] (5 Marks) Paired / Slope Chart

The measure is staff on duty (higher is better). Values were recorded before and after a process change at 6 departments. set.seed(1282265)

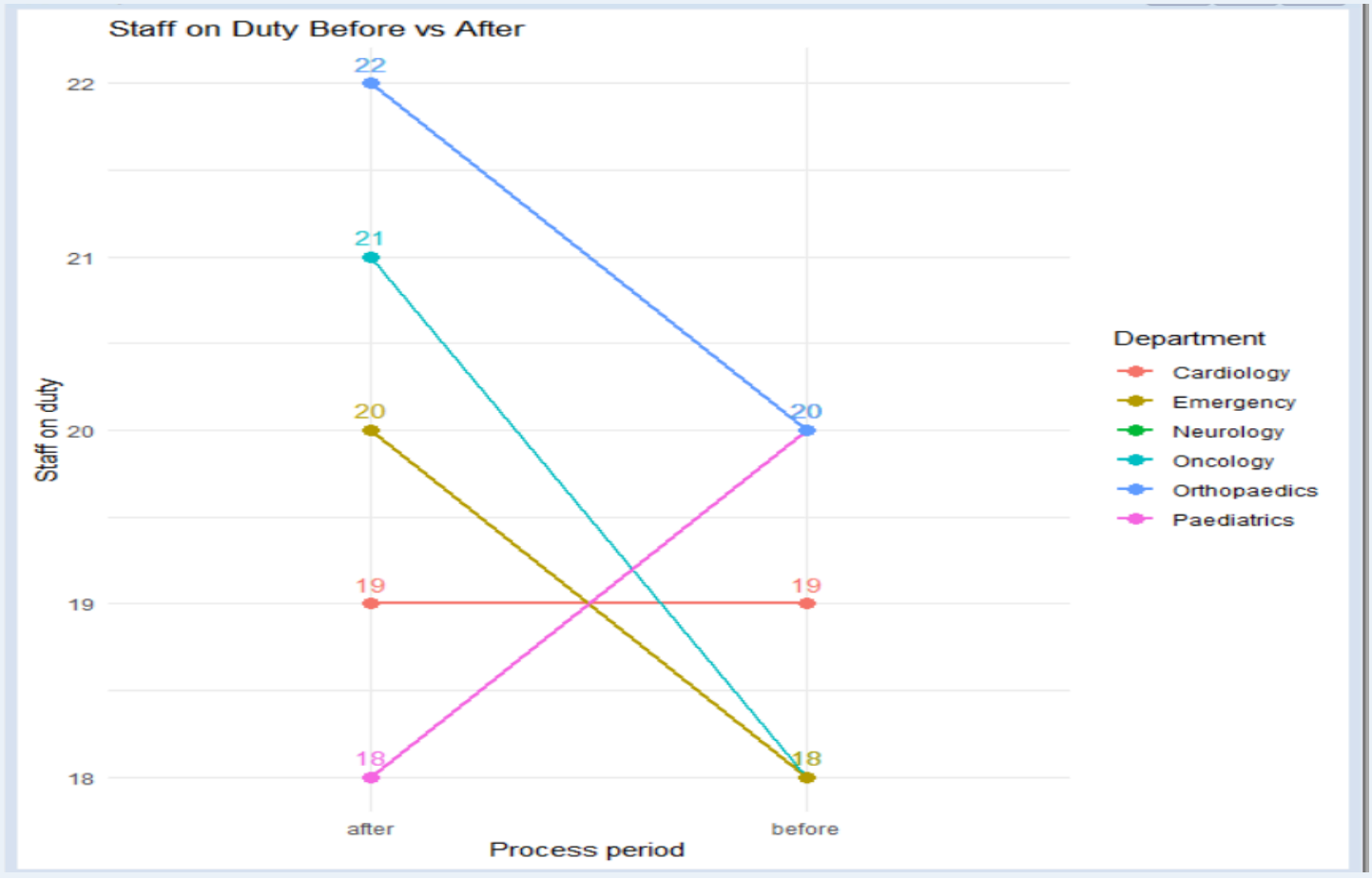
```
entity <- c("Paediatrics", "Cardiology", "Oncology", "Emergency", "Neurology", "Orthopaedics")
n <- 6
before <- round(rnorm(n, 18, 2.0), 0)
change <- round(c(-1, -1, 1, 1, 1, 1) * runif(n, 0.4, 1.4) * 2.4, 0)
change[2] <- 0.0 # one of them was almost unchanged
after <- before + change
df <- data.frame(entity, before, after)
```

Tasks:

1. Draw a paired slope chart (two points per entity joined by a line, geom_line(aes(group = entity))) with labelled endpoints. [1.5] 2. Classify each entity as improved, worsened or flat (flat means the absolute change is below 1% of the mean 'before' value = 0.180) and report the counts. [1]

3. Quantify the largest change (entity, size, and percentage of its 'before' value). [1]

4. Run a paired t-test (t.test(after, before, paired = TRUE)) and report t, p and the mean change; give one recommendation. [1.5]



Q6 [Bubble Charts] (5 Marks) Bubble Chart

Compare 8 genres on marketing spend (Rs lakh) (x), watch time (hours per user per month) (y) and subscribers (lakh) (bubble size). set.seed(1282349)

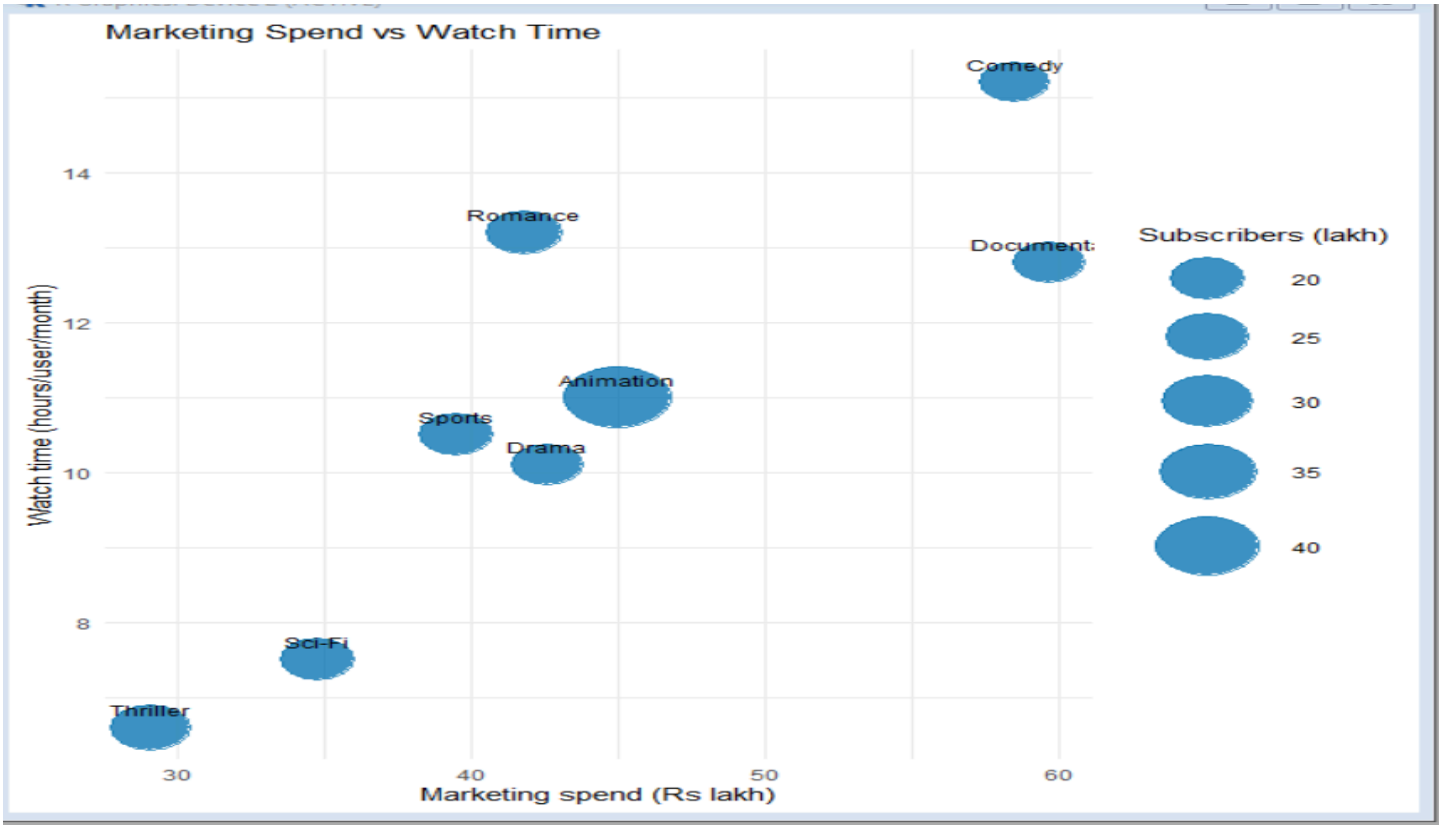
```
entity <- c("Sci-Fi", "Sports", "Romance", "Documentary", "Animation", "Thriller", "Drama",
"Comedy") n <- 8
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.0 * (0.90 * z + 0.40 * rnorm(n)), 1)
```

subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)
subs_lakh[5] <- round(18.0 * 2.4, 1); spend_lakh[5] <- round(45.0); watch_hrs[5] <- round(11.0) # Animation has an unusual subs_lakh
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)

Tasks:

1. Draw a bubble chart with geom_point(aes(size = ...)) and scale_size_area(), labels on every bubble, and a legend for size. [1.5] 2. Report the correlation between the x and y variables. [0.5]

3. Name the entity that is an outlier on the size variable even though its x/y position is near average, and show it with a z-score of size. [1.5] 4. State how many times larger its size value is than the median size, and explain why a plain scatterplot would have hidden it. [0.5] 5. Give one recommendation for the Content Head. [1]



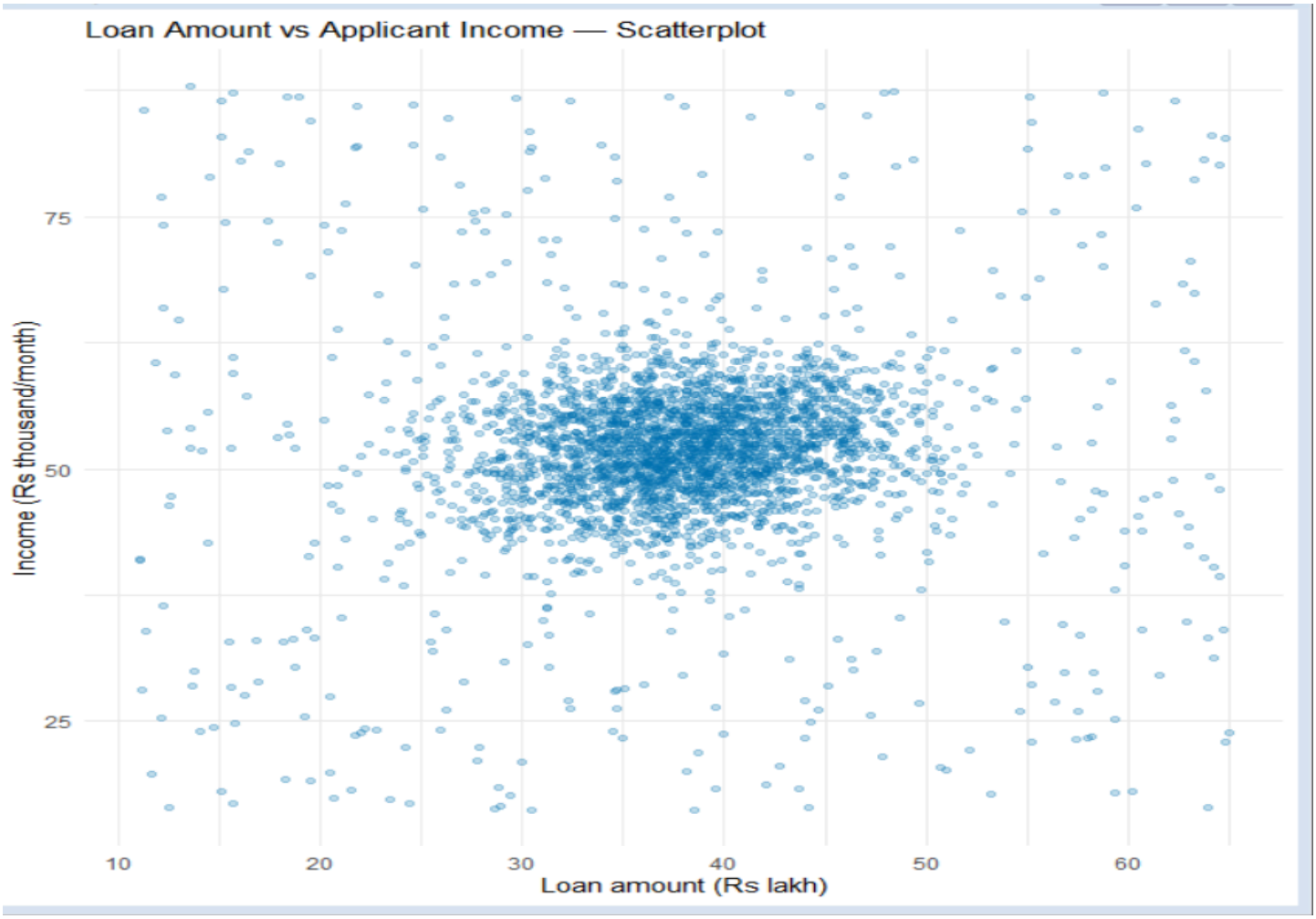
Q7 [Hexbin Plots] (5 Marks) Hexbin Plot

A large paired dataset: loan amount (Rs lakh) (x) and applicant income (Rs thousand per month) (y), 3000 records. set.seed(1282371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4), runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 + 0.78 * (loan_lakh - 38.0) / 5.4 + sqrt(1 - 0.61) * 7.2 * rnorm(n), runif(n, 16.0, 88.0)), 1) df <- data.frame(loan_lakh, income_k)

Tasks:

1. Draw a hexbin plot (geom_hex(bins = 25) or the hexbin package) with a count legend and a viridis scale; also draw the plain scatterplot of the same data. [1.5]

2. Using a 10 x 10 grid (table(cut(x, 10), cut(y, 10))), find the busiest cell: give its x range, y range, count and share of all points. [1.5] 3. Report how many distinct (x, y) pairs there are (nrow(unique(df))) and explain why the scatterplot alone would mislead. [1] 4. Give one recommendation for the Zonal Manager. [1]



Q1. Scatterplot: Weekly Sales vs Footfall

1. Interpretation of the scatterplot

The scatterplot with a linear regression line and LOESS curve is used to analyse the relationship between weekly sales and customer footfall. The plot helps identify the overall trend, possible nonlinear patterns and unusual observations.

2. Correlation and regression analysis

The correlation coefficient is 0.3936, indicating a moderate positive relationship between sales and footfall.

The linear regression equation is:

$$Footfall = 500.0319 + 4.6481(Sales)$$

The slope of 4.6481 indicates that a one-unit increase in sales is associated with an estimated increase of approximately 4.6481 visitors in footfall, according to the fitted model.

The linear model has an R^2 value of 0.1549, meaning that sales explain approximately 15.49% of the variation in footfall.

3. Outlier identification

The identified outlier is Row 18, with sales of 97.5 and footfall of 271. Its standardised residual is −4.1911. The large negative residual indicates that its observed footfall is substantially below the value predicted by the linear regression model.

4. Linear and quadratic model comparison

Model	R-squared
Linear model	0.1549
Quadratic model	0.1606

The quadratic model provides a small improvement in R^2. However, the improvement is limited, indicating that adding a quadratic term does not substantially improve the model's explanatory power.

5. Recommendation to the Sales Director

The Sales Director should investigate the unusual observation in Row 18 and verify its data accuracy. Since the relationship between sales and footfall is only moderately positive, other factors such as pricing, promotions and product availability should also be considered.

Q2. Correlogram: Telecom Measures

1. Correlation matrix and heatmap

A correlation matrix measures the strength and direction of relationships between telecom variables. The correlogram represents these correlations using colours and numerical values, making strong positive, strong negative and weak relationships easier to identify.

2. Two strongest correlations

The two strongest distinct positive correlations are:

Variable pair	Correlation
Users (lakh) and ARPU	0.72
Drop rate and churn percentage	0.70

The correlation of 0.72 indicates that user count and average revenue per user (ARPU) move positively together in this dataset. The correlation of 0.70 indicates that higher call drop rates tend to be associated with higher customer churn.

3. Weakest correlation

The weakest relationship is between tower density and ARPU, with a correlation of approximately −0.01. This value is very close to zero, indicating almost no linear relationship between these variables in the dataset.

This suggests that tower density alone may not explain changes in average revenue per user.

4. Recommendation to the Network Head

The Network Head should prioritise reducing call drop rates because they show a strong positive association with customer churn. The relationship between user count and ARPU should also be examined to understand revenue patterns. Tower density should not be used alone to predict ARPU.

Q3. PCA and t-SNE: Destination Group Analysis

1. Principal Component Analysis (PCA)

PCA reduces the dimensionality of a dataset by transforming correlated variables into a smaller set of principal components while retaining as much variation as possible.

The percentage of variance explained by each component is:

Principal component	Variance explained
PC1	38.54%
PC2	19.26%
PC3	16.18%
PC4	10.69%
PC5	8.87%
PC6	6.45%

PC1 and PC2 together explain 57.80% of the total variance. The first four principal components explain 84.67%, exceeding the required 80%. Therefore, four components are sufficient to retain at least 80% of the dataset's variation.

2. Largest loading on PC1

The largest absolute loading on PC1 belongs to complaints, with a loading of approximately 0.5400. This indicates that complaints contribute strongly to the first principal component relative to the other variables.

The PC1 versus PC2 scatterplot helps visualise similarities and differences among Groups A, B and C.

3. Group separation using R-squared

Component	R-squared
PC1	0.5827
PC2	0.0202

The group variable explains approximately 58.27% of the variation in PC1 scores but only 2.02% of the variation in PC2 scores. This indicates that group differences are much more evident along PC1 than PC2.

4. PCA versus t-SNE

PCA is a linear dimensionality-reduction technique that preserves overall variance and provides interpretable component loadings. t-SNE is a nonlinear technique that focuses on preserving local neighbourhood relationships, helping reveal clusters in high-dimensional data.

However, distances and cluster sizes in a t-SNE plot should be interpreted cautiously because the axes do not directly represent the original variables or explained variance.

5. Recommendation

The destination analysis should use PCA to identify the major factors influencing destination characteristics and reduce the number of

variables. The t-SNE plot can be used as a complementary visualisation to explore possible clusters among destination groups.

Q4. Network Analysis: Parcel Delivery Hub Network

Important: Your Q4 code did not execute because the igraph package is not installed. The following theory can be used, but the numerical network results must be generated before completing the result-specific parts.

1. Network visualisation

A network graph represents cities as nodes and delivery connections as edges. Node size represents degree, while edge width represents connection weight. Cities with larger nodes have more direct connections and may serve as important delivery hubs.

2. Degree and betweenness centrality

Degree centrality measures the number of direct connections a city has. Betweenness centrality measures how frequently a city lies on the shortest paths between other cities. A city with high values for both measures may be important for direct connectivity and the movement of parcels through the network.

The most central city should be identified by comparing the degree and betweenness values produced by the program.

3. Network structure

The number of edges indicates the total delivery connections in the network. Network density measures the proportion of existing connections among all possible connections. The number of connected components indicates how many separate groups of interconnected cities exist.

4. Effect of removing the central city

Removing the most central city helps evaluate the network's resilience. If the number of connected components increases, the removal has divided the network into more disconnected groups. This indicates a possible risk to delivery continuity.

5. Recommendation to the Operations Head

The Operations Head should identify highly central cities and ensure alternative delivery routes are available. Maintaining backup routes and distributing delivery capacity across multiple hubs can reduce dependence on a single city.

To generate the missing results, run this command in R:

```
install.packages("igraph")
```

Q5. Paired Data Analysis: Staff Before and After

1. Slope chart interpretation

A slope chart compares staff levels before and after a process change across six departments. Upward lines indicate an increase in staff levels, downward lines indicate a decrease, and horizontal lines indicate no change.

2. Classification of changes

Department	Before	After	Change	Status
Paediatrics	20	18	−2	Worsened
Cardiology	19	19	0	Flat

Oncology	18	21	+3	Improved
Emergency	18	20	+2	Improved
Neurology	20	22	+2	Improved
Orthopaedics	20	22	+2	Improved

The results show that four departments improved, one remained flat and one worsened.

3. Largest change

Oncology shows the largest change, increasing from 18 to 21. The absolute increase is 3 staff members.

Percentage change:

$$\frac{21-18}{18}\times100=16.67\%$$

Therefore, Oncology recorded a 16.67% increase.

4. Paired t-test

The paired t-test compares measurements taken from the same departments before and after the change.

- t-statistic: 1.5575
- Degrees of freedom: 5
- p-value: 0.1801
- Mean change: 1.1667
- 95% confidence interval: −0.7589 to 3.0922

Since the p-value is greater than 0.05, the change is not statistically significant at the 5% significance level. The observed average increase does not provide sufficient evidence of a statistically significant overall change.

5. Recommendation

Management should examine why Paediatrics experienced a decrease and identify the factors associated with improvement in Oncology and other departments. More observations should be collected to determine whether the overall increase is consistent and statistically meaningful.

Q6. Bubble Chart: Marketing Spend vs Watch Time

1. Bubble chart interpretation

A bubble chart displays marketing spend on the x-axis and watch time on the y-axis. Bubble size represents the number of subscribers. This visualisation compares three variables simultaneously and helps identify relationships and unusual observations.

2. Correlation analysis

The correlation between marketing spend and watch time is 0.8647, indicating a strong positive relationship. In this dataset, higher marketing expenditure tends to be associated with greater watch time.

3. Subscriber-size outlier

Animation is the subscriber-size outlier, with 43.2 lakh subscribers. Its standardised z-score is 2.4119, showing that its subscriber count is considerably higher than the dataset average.

Animation's subscriber count is approximately 2.14 times the median subscriber count.

4. Why a plain scatterplot may be misleading

A plain scatterplot displays only marketing spend and watch time. It does not represent subscriber count through bubble size, so the unusually high subscriber count for Animation could remain unnoticed. The bubble chart reveals this additional information.

5. Recommendation to the Content Head

The Content Head should investigate why Animation attracts a much larger subscriber base than the other genres. The strong positive relationship between marketing spend and watch time is worth examining, but it does not establish that marketing spend alone causes higher watch time. Subscriber count, genre preferences and content quality should also be considered.

Q7. Hexbin Plot: Loan Amount vs Applicant Income

1. Hexbin plot interpretation

A hexbin plot divides a scatterplot into hexagonal bins and represents the number of observations in each bin using colour intensity. It is useful for large datasets because overlapping observations become easier to identify.

The dataset contains 3,000 records, with 2,895 distinct loan-income pairs.

2. Busiest grid cell

The busiest cell in the 10 × 10 grid has the following values:

Measure	**Result**
Loan amount interval	(32.7, 38] lakh
Income interval	(44.9, 52] thousand/month
Number of records	419
Share of all records	13.97%

The busiest cell contains 419 applicants, representing approximately 13.97% of all 3,000 observations. This indicates a concentration of applicants within the specified loan and income intervals.

3. Hexbin plot versus plain scatterplot

A plain scatterplot may suffer from overplotting when many observations occupy similar locations. Overlapping points can conceal the density of observations and make patterns difficult to interpret. A hexbin plot summarises these overlapping observations and highlights high-density regions.

4. Recommendation to the Zonal Manager

The Zonal Manager should examine the high-density loan and income ranges when planning loan-processing resources and evaluating common applicant profiles. However, loan approval decisions should also consider repayment capacity, credit history and other relevant criteria rather than relying on the concentration of applicants alone.

Code of Conduct: I certify that this work is my own. Signature: AKILA S